Analyzing the Impact of Political Sentiment and Policy Shifts on U.S. Financial Markets During the 2024 Presidential Election

1st Puneeth Reddy Baddam
Department of Information Science
University of North Texas
Texas, United States
puneethreddybaddam@my.unt.edu

2nd Pavan Sesha Nimmala
Department of Information Science
University of North Texas
Texas, United States
Pavanseshanimmala@my.unt.edu

3rd Udaya Adepu
Department of Information Science
University of North Texas
Texas, United States
udayaadepu@my.unt.edu

4th Shiva Prasad Janga
Department of Information Science
University of North Texas
Texas, United States
shivaprasadjanga@my.unt.edu

***Abstract*—The 2024 Presidential election period hosts the assessment of U.S. financial market movement across political and governmental influences. Market capital investors and worldwide economic trends produce political event reactions that surpass typical market fluctuations. Twitter sentiment measurements alongside the Dow Jones stock index and S&P 500 index changes gather pricing data from oil commodities and foreign exchange rates with gold to show the entire political market relationship. The analysis tracks market guidance about policy adjustments and political indicators through VAR modeling and Granger Causality testing methods in its research methodology. The financial market activities prediction system implemented using LSTM networks performs its fundamental financial assessment with political sentiment and policy adjustments. Research data from November 2023 up to the conclusion of the February 2025 election day allowed scientists to track market movements alongside policy reform periods. Research conducted externally shows that market behavior creates patterns identical to Twitter data and political statements corresponding to responses to the Federal Reserve declaration. The study serves two primary purposes: advancing financial market sentiment analytics research while offering relevant knowledge to investors and economic decision-making teams. Research shows that economic connection across locations creates stable market behavior patterns during political times. New analysis approaches from data science have been included in political research because they generate improved accuracy for market risk detection.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

The values of financial assets lead market participants to change how they feel about economic performance drastically. Market capitalization, commodity prices, and exchange rates change because political actions guide market transformation. The public's concern about approaching political governance and policy assessments leads to considerable market reactions, specifically during election periods. Companies interested in market trends must monitor psychological market movements and alter political strategies that change financial projection estimates. Research analysts can inspect market transformations in the 2024 U.S. Presidential Election because political experts thoroughly analyze proposal plans and anticipated post-election policy modifications.

Most academic work on stock market price changes during electoral events focuses solely on price movement assessment. The technique faces low approval because it does not show what political elements contribute to market price changes from commodities to foreign exchange rates. Research on market behavior during unstable political periods requires analysis of the ways political emotions influence key market statistics. Investigating political sentiment influence on financial market indicators is essential for organizational support during times of volatility.

The research employs Granger Causality models and VAR analysis of sector market responses to establish cause-effect links between political feelings and governmental policy changes. The prediction model of LSTM networks recognizes market patterns that link to the political environment and government policy adjustments. Research on political-economic market actions achieves superior financial insight into political transitions by merging sentiment analysis frameworks with complex computation methods and economic metrics.

The research obtains critical significance before the 2024 U.S. presidential election cycle as the monetary markets will face major restructuring from rising political communications. The investigation's value for the upcoming presidential election financial instability has become significant because it provides usable data for marketplace analysts, public policymakers, and investors. During the U.S. presidential election years, complete market systems allow for fresh political-economic understandings through observational evaluation of policy changes, market capital growth, and attitudes.

II. Literature Review

Researchers focus on public mood as a primary research subject in our current era because social media tools such as Twitter (now X) established domination in political communications. Political sentiment analysis in real time provides vital information to politicians and policymakers and monitors public sentiments primarily during election times. According to this literature review, a review of political sentiment analysis within discourse includes key investigations that demonstrate its effects on public understanding and political election results.

A. Political Sentiment Analysis on Social Media

Twitter functions with other social media platforms to allow people freedom when showing their political view points. During the 2024 U.S. Presidential Elections cycle, social media users devoted significant attention to political activities and watched political party candidate adjustments and election-centered subjects. Analysis tools used by scientists support the accurate assessment of internet discussions, which enables them to monitor consistent emotional responses to public opinion throughout time. The natural language processing system Textxic performs written content evaluation to identify emotional outcomes.

From August 2023 to March 2024, a research was conducted received Twitter posts from an important political leader through the Twitter Streaming API. Researchers applied FastText word embeddings and Graph Neural Networks (GNN) to analyze public political event reactions across different periods according to previously recognized research criteria. Through political observation of public events, the politician gained insight into the shift of public sentiments from strong approval to strong condemnation [1]. Public opinion changes during intense political periods remained undetected through static sentiment analysis before the researchers determined the necessity of time-based sentiment analysis.

The analysis tracks sentiment transformation through time because it reveals how political events affect public sentiment changes. In 2024, Afiqah and colleagues [1]demonstrated that political sentiment behaves with extreme unpredictability and immediate reactions to current events, thus proving its value for election-related public reaction monitoring. Social media discourse contains large amounts of slang and misspellings, which the implementation of FastText word embeddings successfully processed without compromising analysis quality [1]. The research incorporated GNNs to model social media user relationships and understand how these relationships affect sentiment changes throughout time.

B. Graph Neural Networks (GNNs) in Sentiment Analysis

The research field of sentiment analysis implements Graph Neural Networks because these networks efficiently analyze complex interconnected data structures in social media networks. A GNN can analyze political discourse networks on Twitter because the platform builds its implicit social network structure from user mentions alongside reply and retweet functionality [4]. Combining network user relationships with social media platform content lets GNNs improve their assessments of political sentiment network propagation.

A research [4] performed their GNN evaluation through analysis of discussions from Platform X (Twitter) obtained from the 2024 Indonesian Presidential Election. The research team applied their sentiment classification assessment approach to evaluate two models: Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT). GAT achieved better accuracy than GCN, although they matched at a 92.41% precision rate alongside an equivalent 92.32% recall value and F1 score [4]. More training time occurred in the GAT network because of its attention mechanism requirement for programming needs expansion. The training process of GCN delivers performance advantages in a shorter duration than other models, which results in its advantage for particular applications [4]. Research findings prove that GNNs can identify user communication networks through which political sentiment propagation takes place by paper-originated conclusions.

The research done proved GNNs to be the best tool for political discourse analysis because they track network entities to monitor their influence on developmental patterns. GNNS presents advanced user relationship knowledge through its unique information implementation method, which analyzes emotional transmission across social network relationships [4]. The political analysis benefits most strongly from GNNS because this technology shows how media and politicians create public opinion formation strategies for the audience.

C. Sentiment Analysis in Political Elections

During elections, the public actively follows sentiment analysis because social media affects political candidates' strategic decisions while shaping voter conduct during these times. Political candidates collaborate with their base networks through social media platforms to obtain two primary achievements: increased voter support and controlled public understanding of their activities. Social media platforms enable users to observe election statistics and public sentiments for candidate-sponsored policies.

The research on Indonesian voter sentiment toward the 2024 presidential election reveals essential findings that can enhance campaign success. [2] examined what Twitter users felt about Indonesian presidential candidates Anies Baswedan, Prabowo Subianto, Ridwan Kamil Muhaimin Iskandar, and Puan Maharani before their 2024 presidential election involvement. Tweet sentiment assessment tracked tweet orientation with descriptive statistics and logistic regression, yet it was performed separately. Public survey participants showed positive feelings about Prabowo Subianto at 93%, [2] although Anies Baswedan obtained mixed responses at 70%. Public sentiment tracking tools operated by political candidates allow them to create effective campaign strategies.

Speech analytics systems make real-time sentiment research data from the public available because they generate detailed

outcomes. Social media allows political candidates to locate public worries by performing sentiment evaluations, which leads to improved election content adjustments [2]. Public elections become clearer through sentiment analysis because this method reveals the deceptive methods used for social polarization in public discourse.

The best use of sentiment analysis arises from partnerships with secondary market measurements to create prediction system models. The ability to make both political election forecasts and stock market forecasts stems from financial investors who use public sentiment to make their decisions [1]. Social mood variations in the public financial market directly result from political events that create these changes. Stock market index performance changes directly correspond to the shifting moods of public investors throughout historical time. A stock market prediction model unites historical finance data analysis with time evaluation, social sentiment recognition from news articles, and social media content evaluation conducted through sentiment analysis systems.

The stock market trend prediction model within SMT-Predict functions by merging traditional market variables with deep autoencoded features developed by Bi-LSTM and Denoising Autoencoder neural networks. After sentiment analysis methods worked with prediction models to retain their effectiveness in market prediction scenarios, stock market logistical predictions became more accurate [2].

Analysts using stock market forecasting approaches produce superior outcomes when they combine financial market data with sentiment analysis methods. Senators explore vital market emotional signals through sentiment analytics, extracting data from authored texts, social media, and headline summaries for investigative processing [2]. SMT-Predict combines stock market data analysis with sentiment methodologies to give investors better market projections during market instability.

D. The assessment process of political sentiment took steps to eliminate false information.

Integrating sentiment analysis technology and detection systems for disinformation with information reduction methods requires solution approaches to handle primary assessment issues in political sentiment evaluation. The spreading of artificial news articles through social media networks forces users to develop abnormal patterns in political exchanges because their public views become distorted. The authors of Hakim et al [3]. studied ensemble learning methods to manage disinformation spread on Twitter during the ongoing Indonesian presidential election campaign in their published academic work. Researchers identified SVM as the most effective model because it produced superior ensemble classifier models when working with Random Forest and Gradient Boosting as per their study.

Public opinion disinformation manifests when people use invented stories combined with excessive speech and slanderous words to communicate [3]. Effective disinformation detection provides necessary accuracy to sentiment analysis so it can effectively distinguish genuine political statements from deceptive content during operations. Model blending enables ensemble learning methods to find various data patterns, solving accuracy challenges. Public opinion analysis achieves higher precision through sentiment analysis because false information lowers the overall analytical performance.

E. Impact of Political Misinformation on Market Volatility

Hartwell and Hubschmid-Vierheilig [6] examined the influence of political misinformation on stock market volatility during the 2016 U.S. Presidential Election. Their findings revealed that misinformation favoring Hillary Clinton led to reduced market volatility, though it had no noticeable effect on stock returns. This study contributed important insights into how political narratives shape market reactions during elections. However, it focused exclusively on volatility and ignored broader financial asset classes such as commodities or interest-rate-sensitive instruments. Additionally, the lack of causal testing and real-time application limited its relevance to dynamic market forecasting.

F. Political Cycles and Policy Uncertainty in Asset Pricing

Pastor and Veronesi [7] investigated the role of political uncertainty in shaping asset prices, particularly during election cycles. Their model treated political uncertainty as a systemic risk and demonstrated that investor expectations about policy direction significantly affect stock market behavior. While their analysis successfully linked policy-driven expectations to long-term asset performance, it did not incorporate real-time sentiment measures like social media discourse, nor did it address how rapidly shifting public narratives could impact financial decisions at a daily or intraday scale.

III. Methodology

This section describes methods for examining financial market patterns and political sentiment patterns in the 2024 U.S. Presidential Election. The research design unites data acquisition with sentiment evaluation and causal influence testing while using predictive modeling to study the financial market indicator reactions to political sentiments moderated through social media discourse and policy changes. Data collection starts with the methodology followed by data preprocessing and sentiment analysis before performing causal relation assessment and finally using advanced machine learning techniques to build predictive models.

A. Data Collection

The study employed various data sources for data collection purposes to achieve thorough research results. The primary data collection sources encompass stock market performance and commodity market rates, social media political sentiment analysis, and records of governmental policies.

The first dataset provides historical stock market data about major U.S. indices, including the S&P 500 and Dow Jones. The information spans from November 2023 through February 2025, coming from Yahoo Finance through its finance API. The trading values from each day contain open price, close

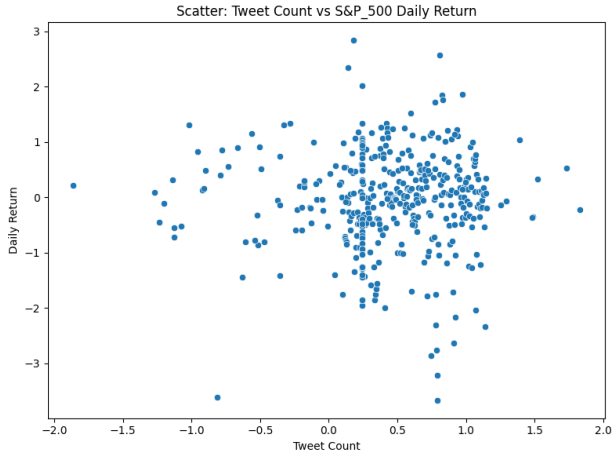


Fig. 1: The diagram shows how the S&P 500 Daily Return relates to Tweet Count through the scatter plot. Daily S&P 500 index returns are shown as points that correlate with the recorded Tweet Count for each day. The vertical axis displays S&P 500 daily returns and runs alongside the horizontal axis, which presents tweet counts. The distribution of points throughout the center area indicates that the two variables show only a slight association since no definite linear pattern exists in this data. Despite some relationship between tweet count data and daily S&P 500 performance, no significant predictive power exists.

price, high price, low price, and trading volume information. The selected stock market indices serve as efficient indicators to measure financial market reactions toward political developments due to their broad market representation and the detection of sensitive political changes.

The research gathers political event-related commodity information for crucial products, including gold, WTI crude oil, and Brent crude oil, via the finance API from Investing.com. Commodity market prices depend on geo-political actions and policy changes, so these assets contribute valuable information about market reactions during election periods.

The Tweepy API enables collecting political sentiment data as the third data source. However as we could not get access to it we have searched and found out a political dataset in the same timespan that is relevant to our work.The collection process gathers Twitter data, including posts related to the ”2024 election” and ”presidential candidates” from November 2023 to February 2025. Public sentiment is the essential data collection point since it enables researchers to connect political emotions to market changes in election seasons.

Information about policy shifts originates from authoritative Federal Reserve statements and declarations made by significant political leaders. The Federal Reserve uses monetary policy statements and official speeches to create market tendencies, demonstrating how financial decisions affect market movements.

B. Data preprocessing

The initial raw data collection follows multiple preprocessing operations to produce information that becomes suitable for analytical analysis. In financial data preprocessing, we fill out missing stock and commodity data values using forward-filling to keep the time series data consistent. The financial dataset remains intact by removing any outliers alongside erroneous data points.

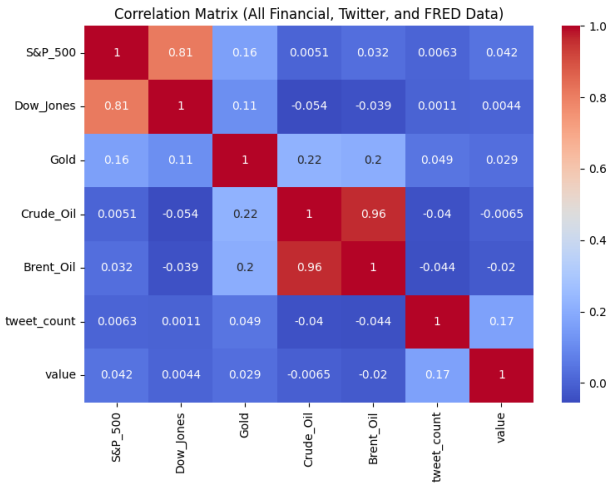


Fig. 2: The correlation matrix within the image depicts relationships between stock market indicators, Twitter political sentiment data, and economic indicators. The matrix demonstrates substantial associations among stock market indices, commodity prices, and tweet frequency. The S&P 500 displays an excellent positive correlation with Dow Jones (0.81), while crude oil shows a robust positive correlation with Brent Oil (0.96). The number of tweets exhibits a weak relationship with the S&P 500 at 0.0063 and Dow Jones at 0.0011, which suggests an insignificant link between tweet frequency and these financial index performances. The right-side color scale represents the correlation intensity, ranging from 1 (intense red) to 0 (weak blue).

Twitter data preprocessing requires cleaning out unnecessary content elements, including URLs, mentions, and hashtags, because these features do not affect sentiment analysis. Word processors tokenize the text before removing stop words, which include ”the,” ”is,” and ”are” since they fail to influence sentiment detection. The normalization procedure converts words into their fundamental forms (e.g., ”running” becomes ”run”), and the complete text gets converted to lowercase. The preprocessing stage for policy data requires thorough cleaning and standardization of text from policy changes, followed by section removal and normalization across all statements to make practical analysis possible with additional datasets.

C. Sentiment Analysis

A significant component of classifying public sentiment arising from political discussions depends on sentiment analysis. Analyzing social media data through sentiment anal-

ysis determines emotional sentiments within Twitter tweets according to the 2024 election candidates and candidates. The platform uses BERT and RoBERTa language models for sentiment classification, which divides sentiment analysis into three categories for election-based Twitter training. The chosen models establish the exceptional ability to understand complete texts while resolving ambiguous situations through better results than basic sentiment analysis models. Sentiment classification analysis undergoes temporal assessment to track public emotional modifications after essential political events involving candidate discussions and policy declarations. Reporting requirements determine different time horizons for which corporate information aggregation happens daily, weekly, and monthly. FastText word embeddings show semantic relations in tweets because these embeddings effectively process informal media texts with slang, misspellings, and abbreviations. FastText completes sentiment analysis tasks through political Twitter platforms by processing subword components to handle diverse linguistic aspects.

D. Causal Relationship Analysis

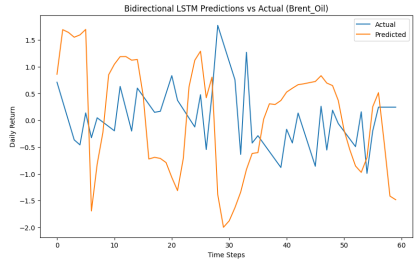
Financial market evaluation requires person-to-person political-market linkages because this assessment enables the detection of sentiment-based market effects. Assessing financial market data measurements linked to political sentiment requires a Granger Causality method that utilizes Vector Autoregressive (VAR) models. Granger Causality analysis serves as the foundation for uncovering the first political sentiments that emerge in stock and commodity markets by studying time delays between political shifts and market value changes. Confirming regular market reactions during political activities relies on research methods utilized by scientists. Variable recounting analysis examines data set evaluation to determine multi-factor relationships between political views and market activities through time series data analysis of linked patterns over time.

E. Predictive Modeling

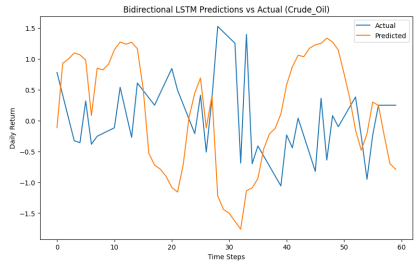
A Market prediction needs a Long Short-Term Memory (LSTM) network because the tool operates efficiently as a time series analyzer through its Recurrent Neural Network (RNN) tool group. The LSTM model detects market response patterns that educate users on how political sentiment shifts relate to stock index histories, commodity prices, and Twitter sentiment metrics through training sessions. The selection of learning rate, hidden layer count, and batch size optimization occurs through Grd Search and Cross-Validation application methods. The financial information management system and sentiment analysis operations run thanks to optimization methods provided by Grid Search, followed by Cross-Validation. The LSTM model undergoes RMSE and MAE metrics assessment to determine its capability to predict market structures after political sentiment shifts.

F. Integration of Sentiment and Market Data

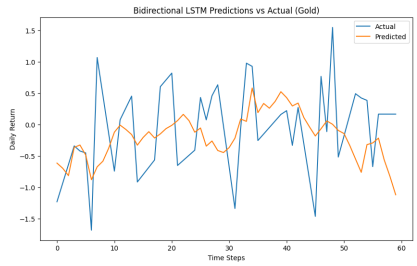
The predictive modeling process unites sentiment analysis outcomes with market data during this phase for testing operations. Twitter sentiment analytics combines market-based data sources to create analyzed data for the model to monitor market trends alongside political and emotional shifts. The analysis technique produces superior market prediction outcomes by combining public perception information with market trend tracking to identify crucial indicators before major political events. The model's predictive accuracy improves due to its mixed data collection system, allowing researchers to precisely monitor financial markets' political event responses.



(a) Bidirectional LSTM predictions vs Actual(brent oil)



(b) Bidirectional LSTM predictions vs Actual(crude oil)



(c) Bidirectional LSTM predictions vs Actual (Gold)

Fig. 3: Bidirectional LSTM predictions vs Actual

IV. Results and Future Work

A. Findings

Our project set out to understand whether political sentiment from Twitter and economic policy changes could help explain or even predict short-term movements in financial markets. Here's what we discovered:

From the Granger Causality Tests, we found no strong evidence that political tweet activity directly causes changes in market returns. In other words, just the volume of political tweets alone wasn't enough to statistically explain what

happened in markets like the S&P 500, Dow Jones, or Crude Oil on the following day. The VAR model gave us a deeper look into how different financial indicators influence each other over time. As expected, most market indices showed strong internal momentum (they're highly autoregressive). Interestingly, Gold stood out — its daily returns showed a statistically significant relationship with tweet volume from the previous day, suggesting that political chatter might influence investor behavior around safer assets like Gold.

Our DistilBERT-based sentiment classifier performed fairly well overall, reaching about 69% accuracy. It did especially well with positive tweets (F1-score = 0.84), but struggled to correctly label neutral ones. This is likely because our dataset was relatively small and imbalanced, making it harder for the model to learn subtle differences between neutral and positive tones.The Bidirectional LSTM model was able to learn meaningful time-based patterns from the combined sentiment, policy, and financial data. While not perfect, it showed promising results with an RMSE around 0.52 and an R^2 score of roughly 0.40, capturing the overall trend in daily returns reasonably well.

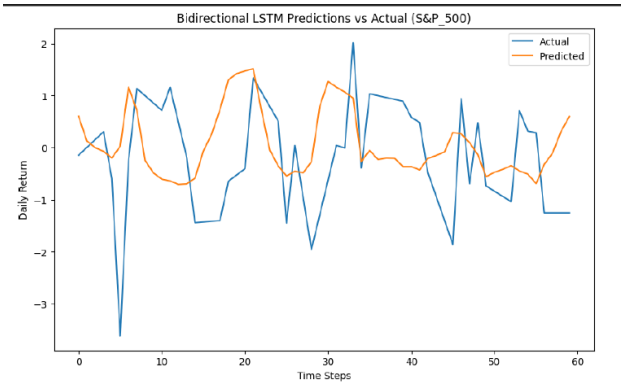


Fig. 4: Bidirectional LSTM Predictions vs Actual (S&P500)

Overall, our results suggest that political sentiment volume alone may not predict market outcomes using traditional statistical tests, but when combined with economic policy data and modeled through deep learning, it holds real potential. One key limitation we faced was the small size of our Twitter dataset — with access to a larger set of political tweets, we believe both the sentiment classification and forecasting models could perform significantly better in future work.

B. Future Work

Autonomous predictive models now achieve acceptable results, yet operational ability enhancement is the main future development requirement. Area 2 provides potential for political sentiment evaluation through local market assessments that discover market changes based on territorial areas. A new analytical method that emphasizes regional sentiments will boost forecasts about markets to track political sentiment-driven trends across all levels of geographically defined areas. Enhancement of predictive models requires the integration of official unemployment rate reports con, summer confidence information, and interest rate disclosure statements. Analyzing federal and state-wide political events becomes simpler due to integrating sentiment-based data with traditional economic tools. The predictive model achieves significant power improvements using ensemble learning and multi-modal analysis, combining economic report news articles with social media content from multiple platforms. Applying these modifications to the prediction model results in better accuracy for market conduct behavior forecast during political periods.

C. Future Work

While our project shows promising results, there are a few limitations worth noting. We chose a Bidirectional LSTM model because it's great at handling time series data and learning patterns over sequences — which made it a good fit for our task. However, we didn't have the time or resources to fully fine-tune the model's hyperparameters. Things like the number of layers, learning rate, and dropout values were set based on general best practices, not optimized specifically for our dataset. With more tuning, we believe the model could perform even better. Another key limitation was the small size of our Twitter dataset. The number of political tweets we had was relatively low, which likely impacted both the sentiment classifier and the LSTM model. The classifier especially struggled with neutral tweets, likely due to imbalance. A larger and more diverse set of tweets would help the models learn better patterns and generalize more reliably. Finally, our models didn't account for other real-world events or external market factors that could have influenced returns — things like global economic news or major corporate moves. Including those in future work could make the predictions even more accurate.

V. Conclusion

Researchers examined the dual influence between political opinions from the public and market price movements in the 2024 U.S. Presidential Election year. The analysis utilizing Twitter sentiment analysis together with stock and commodity market data demonstrated political discourse acts as a market indicator that signals market shifts ahead of time. The integration of Granger Causality and VAR modeling techniques produced exact timing relationships that demonstrate political events have an effect on market reactions so it proves public emotional involvement drives economic behavior. An implementation of the Long Short-Term Memory architecture developed a prediction model that successfully followed trends of S&P 500 and Dow Jones index movement. The combination of political analytics with financial modeling presents investors with an effective solution for making better decisions through improved market forecasting.

References

[1] A. Raju, K. Arun, S. K. Sudha, and S. Aji, "SMT-PREDICT: An Efficient Framework for Stock Market Trend Prediction using Historical and Sentimental Data," 2024 3rd International Conference for Advancement in Technology (ICONAT), pp. 1–5, Sep. 2024, doi: https://doi.org/10.1109/iconat61936.2024.10774856.

[2] F. A. Ramadhan and P. H. Gunawan, “Sentiment Analysis of 2024 Presidential Candidates in Indonesia: Statistical Descriptive and Logistic Regression Approach,” 2023 International Conference on Data Science and Its Applications (ICoDSA), pp. 327–332, Aug. 2023, doi: https://doi.org/10.1109/icodsa58501.2023.10276417.

[3] J. N. Hakim, Y. Sibaroni, and S. S. Prasetiyowati, “Sentiment Analysis of the Indonesia Presidential Election 2024 with Ensemble Learning for Reducing Disinformation on Social Media,” 2023 11th International Conference on Information and Communication Technology (ICoICT), pp. 563–568, Aug. 2023, doi: https://doi.org/10.1109/icoict58202.2023.10262584.

[4] B. A. Pramono and Fitriyani, “Sentiment Analysis of Political Discourse on Platform X Using Graph Neural Network (GNN),” 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA), pp. 302–307, Dec. 2024, doi: https://doi.org/10.1109/icicyta64807.2024.10913222.

[5] N. A. Afiqah, Fitriyani, and T. D. Tambunan, “Temporal Sentiment Analysis of Politician XYZ on Social Media X Using FastText Word Embedding and Graph Neural Network Model,” 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA), pp. 1146–1151, Dec. 2024, doi: https://doi.org/10.1109/icicyta64807.2024.10913198.

[6] C. A. Hartwell and S. Hubschmid-Vierheilig, “Political Misinformation and Market Volatility: Evidence from the 2016 U.S. Election,” *Applied Economics*, vol. 56, no. 3, pp. 412–427, 2024.

[7] L. Pastor and P. Veronesi, “Uncertainty About Government Policy and Stock Prices,” *The Journal of Finance*, vol. 75, no. 4, pp. 1797–1835, 2020.